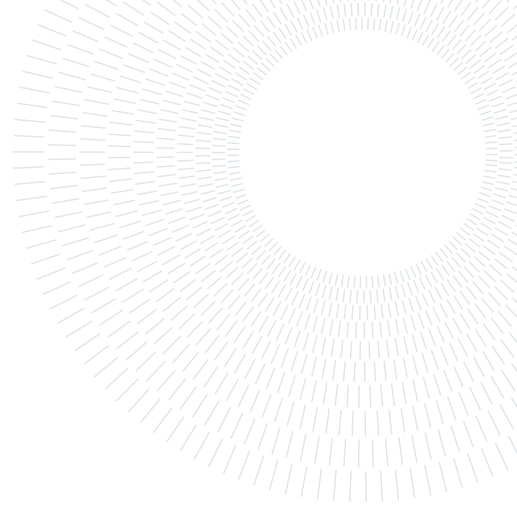




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AEROSERVOELASTICITY OF FIXED AND ROTARY WING AIRCRAFT – A.A. 2025-2026

Workshop 1 - Goland Wing model

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1. Introduction

This report presents a systematic aeroelastic study of the Goland wing, focusing on the characterization of its structural dynamics and stability boundaries. The investigation begins with a **Finite Element Method (FEM)** discretization of the wing's bending and torsion, ensuring numerical convergence through cubic Hermite and linear Lagrange shape functions. Using a **Rayleigh-Ritz modal synthesis**, the uncoupled structural modes are integrated to resolve the coupled natural frequencies induced by the wing's static unbalance.

The core analysis determines the flutter boundary using the $p-k$ **method** and Theodorsen's unsteady aerodynamic theory. This framework is further utilized to perform a sensitivity analysis on physical parameters and evaluate the impact of subsonic compressibility via the **Prandtl-Glauert correction**. Finally, the study explores stabilization strategies through **mass balancing** with tip weights and assesses the aeroelastic implications of control surfaces, identifying critical limits for flutter, control reversal, and static divergence.

2. Task 1: Sensitivity to the number of elements

2.1. Finite Element Formulation

The Goland wing is modeled as a cantilever beam, decoupling its structural dynamics into bending (vertical displacement w) and torsion (twist angle θ).

2.1.1 Bending Model

The bending behavior is governed by the Euler-Bernoulli beam theory. The continuous displacement $w(x, t)$ is discretized using cubic Hermite polynomials as shape functions ($N_i(x)$). This choice ensures C^1 continuity (displacement and slope) across element boundaries, which is required for the 4th-order governing partial differential equation. The elemental mass (M^e) and stiffness (K^e) matrices are

derived using the principle of virtual work, resulting in the standard consistent mass formulation:

$$K_{bending}^e = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}, \quad M_{bending}^e = \frac{mL}{420} \begin{bmatrix} 156 & 22L & 54 & -13L \\ 22L & 4L^2 & 13L & -3L^2 \\ 54 & 13L & 156 & -22L \\ -13L & -3L^2 & -22L & 4L^2 \end{bmatrix} \quad (1)$$

where L is the element length, EI is the bending stiffness, and m is the mass per unit length.

2.1.2 Torsion Model

The torsional behavior is modeled using St. Venant theory. Since the governing equation is of 2nd order, linear shape functions (Lagrange polynomials) are sufficient to satisfy C^0 continuity. The elemental matrices are:

$$K_{torsion}^e = \frac{GJ}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \quad M_{torsion}^e = \frac{I_p L}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad (2)$$

where GJ is the torsional stiffness and I_p is the polar moment of inertia.

2.2. Convergence Study Methodology

To ensure numerical accuracy, a convergence study was performed by varying the number of elements (N_{el}) from 2 to 10. The computed mode shapes (ϕ_{FEM}) were compared against the analytical exact solutions (ϕ_{exact}) for a uniform cantilever beam.

To avoid numerical artifacts at the zero-crossings (nodes) of the higher modes, a global error metric based on the relative L_2 norm was implemented:

$$\varepsilon = \frac{\|\phi_{exact} - \phi_{FEM}\|_{L_2}}{\|\phi_{exact}\|_{L_2}} = \frac{\sqrt{\int_0^L (\phi_{exact}(x) - \phi_{FEM}(x))^2 dx}}{\sqrt{\int_0^L (\phi_{exact}(x))^2 dx}} \quad (3)$$

Prior to comparison, both analytical and numerical modes were normalized to unit tip displacement to ensure consistent scaling.

2.3. Results

The convergence results are presented in Figure 7.

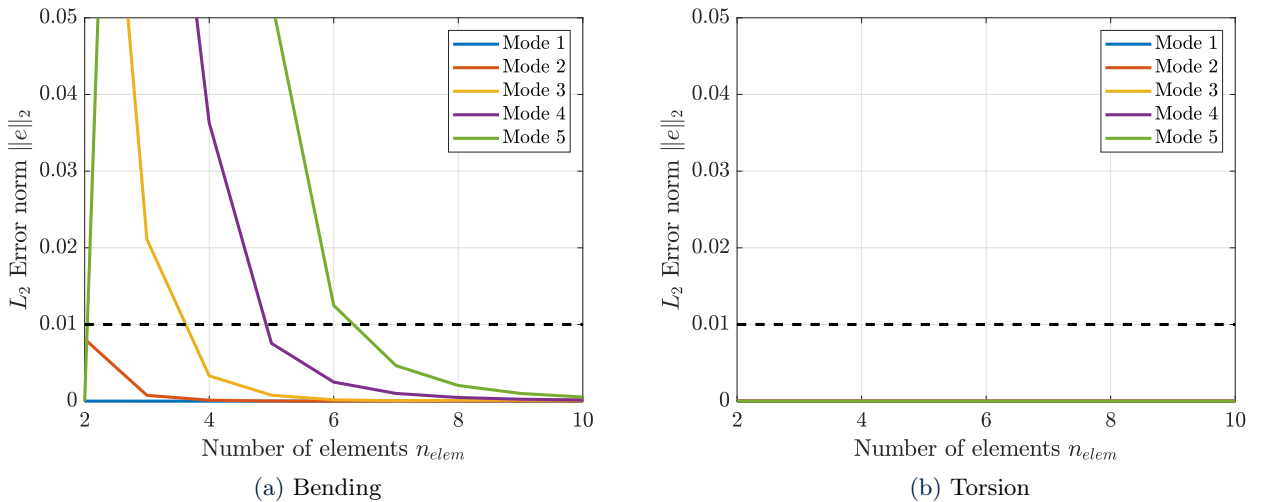


Figure 1: Convergence Analysis

- **Bending:** The use of cubic Hermite shape functions provides rapid convergence due to their C^1 continuity. An error below 1% is achieved with as few as **7 elements**, even for the 5th bending mode. The higher-order polynomials can accurately capture the complex curvature of bending modes.
- **Torsion:** Despite using linear shape functions (which are theoretically lower-order), the convergence is exceptionally fast. The error becomes negligible ($< 10^{-10}$) even for coarse meshes.

Based on this convergence study, a mesh of **10 elements** is sufficient to capture both bending and torsion mode shapes with errors below 1%. However, for subsequent aeroelastic analysis requiring accurate integration of distributed aerodynamic loads, a finer mesh will be employed to ensure precise force distribution along the wing span.

3. Task 2: Convergence of Coupled Modal Frequencies

3.1. Coupled Dynamics Formulation

The dynamic behavior of the Goland wing involves inertial coupling between bending and torsion due to the static unbalance (x_α), which is the offset between the center of gravity and the elastic axis. Since the uncoupled modes computed in Task 1 do not account for this interaction, a Rayleigh-Ritz approach (Modal Synthesis) was employed to compute the coupled natural frequencies.

The total deformation is approximated as a linear combination of the uncoupled mode shapes:

$$w(x, t) \approx \sum_{i=1}^{N_w} q_{w,i}(t) \phi_{w,i}(x), \quad \theta(x, t) \approx \sum_{j=1}^{N_\theta} q_{\theta,j}(t) \phi_{\theta,j}(x) \quad (4)$$

where N_w and N_θ are the number of bending and torsion modes retained in the basis.

Substituting this expansion into the kinetic and potential energy expressions leads to a generalized eigenvalue problem $[\mathbf{K}_{gen} - \omega^2 \mathbf{M}_{gen}] \mathbf{q} = 0$. The stiffness matrix $\mathbf{K}_{gen}$ is diagonal, containing the squared uncoupled frequencies. The mass matrix $\mathbf{M}_{gen}$, however, is populated with off-diagonal coupling terms:

$$\mathbf{M}_{gen} = \begin{bmatrix} \mathbf{I} & \mathbf{M}_{h\alpha} \\ \mathbf{M}_{\alpha h} & \mathbf{I} \end{bmatrix}, \quad M_{h\alpha}^{(i,j)} = \int_0^L \phi_{w,i}(x) \cdot m(x) x_\alpha(x) \cdot \phi_{\theta,j}(x) dx \quad (5)$$

These terms physically represent the torque generated by the inertial force of the offset mass during bending acceleration.

3.2. Convergence Methodology: Richardson Extrapolation

To estimate the asymptotic value of the coupled natural frequencies as the number of basis functions tends to infinity ($N_w, N_\theta \rightarrow \infty$), Richardson extrapolation was employed. This technique exploits the known convergence rate of the numerical method to provide an estimate of the exact solution.

Assuming the modal frequencies converge with second-order accuracy ($p = 2$) with respect to the number of basis functions, the extrapolated frequency f_{ex} is computed as:

$$f_{ex} \approx f_{2h} + \frac{f_{2h} - f_h}{2^p - 1} = f_{2h} + \frac{f_{2h} - f_h}{3} \quad (6)$$

where f_h and f_{2h} represent frequencies computed with basis sizes N and $2N$, respectively.

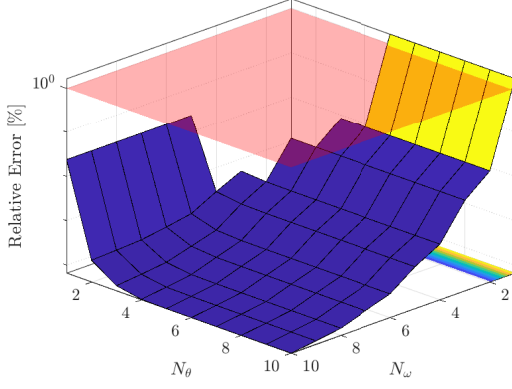
A two-step strategy was implemented: first extrapolating in the bending dimension with fixed torsion, then in the torsion dimension using the intermediate results.

3.3. Modal Identification and Basis Selection

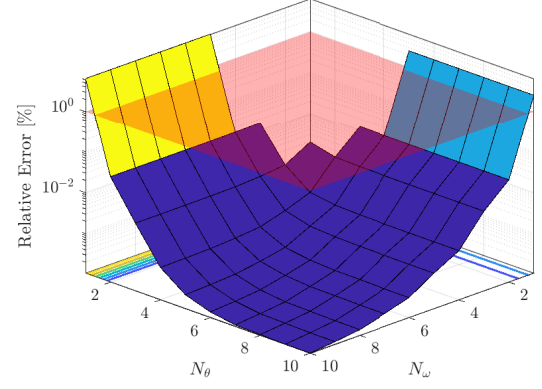
The error surfaces corresponding to the first six coupled modes were analyzed. Based on the dominant deformation mechanism, the modes can be identified as follows:

- **Mode 1:** 1st Bending.
- **Mode 2:** 1st Bending + 1st Torsion.
- **Modes 3-6:** Higher-order bending and torsion modes.

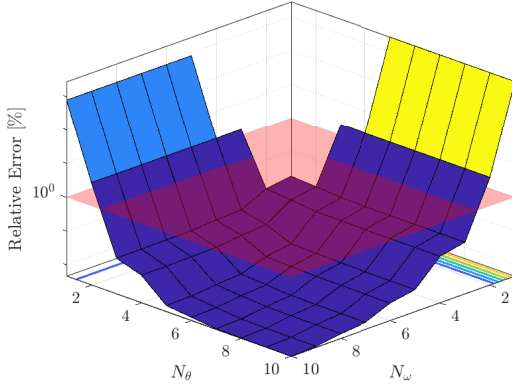
The relative error of the first 6 coupled modes was computed against this extrapolated reference value for basis sizes ranging from 1 to 10 modes. For better visualization, the relative error in the Z-axis is plotted on a logarithmic scale. Additionally, a translucent red plane indicates the 1% threshold; all basis combinations lying below this plane satisfy the accuracy restriction.



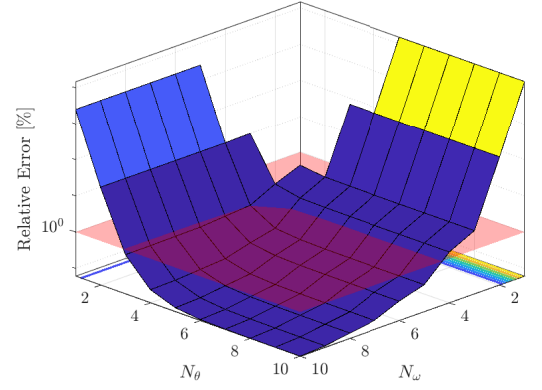
(a) Mode 1



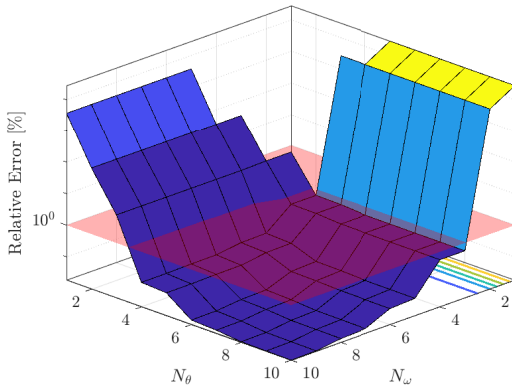
(b) Mode 2



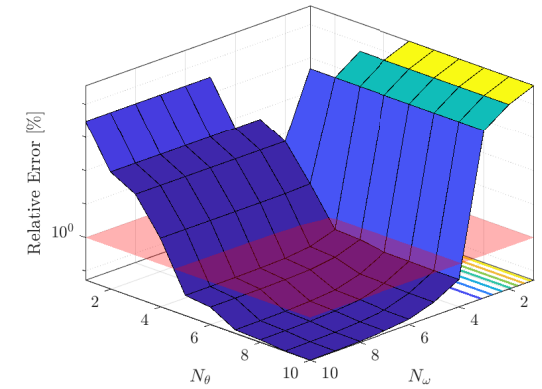
(c) Mode 3



(d) Mode 4



(e) Mode 5



(f) Mode 6

Figure 1: Convergence surface of the relative error for the coupled modal frequencies.

It was observed that higher-order modes require a larger basis set to achieve the target accuracy. To ensure that the error remains below 1% for all the cases when:

$$N_w \geq 5 \quad \text{and} \quad N_\theta \geq 5 \quad (7)$$

4. Task 3: Optimal Modal Basis and Flutter Speed Calculation

4.1. Theoretical Framework: The p-k Method

The aeroelastic stability of the Goland wing is governed by the interaction between the structural dynamics and the unsteady aerodynamic forces. The equations of motion, projected onto the generalized modal coordinates $\mathbf{q}$, are expressed in the frequency domain as:

$$\left[\left(\frac{U}{b} \right)^2 p^2 \mathbf{M}_{gen} + \mathbf{K}_{gen} - q_\infty \mathbf{Q}_{gen}(k) \right] \bar{\mathbf{q}} = 0 \quad (8)$$

where $p = \gamma k + ik$ is the complex eigenvalue (with γ representing the damping decay rate), $q_\infty = \frac{1}{2}\rho U^2$ is the dynamic pressure, and $\mathbf{Q}_{gen}(k)$ is the generalized aerodynamic force matrix.

The aerodynamic loads are modeled using Theodorsen's Strip Theory. The unsteady effects and phase lags are captured by the reduced frequency $k = \omega b/U$ and Theodorsen's Transfer Function $C(k)$:

$$C(k) = \frac{H_1^{(2)}(k)}{H_1^{(2)}(k) + iH_0^{(2)}(k)} = F(k) + iG(k) \quad (9)$$

Since $\mathbf{Q}_{gen}$ is a transcendental function of the unknown eigenvalue (via k), the system is nonlinear. The *p-k method* was implemented to solve this iteratively: for a fixed airspeed U , the reduced frequency k is updated until the assumed frequency matches the imaginary part of the computed eigenvalue (the "Matched Point").

4.2. Optimal Modal Basis Selection

A sensitivity study was conducted to determine the minimal set of proper orthogonal modes required to reduce the error in the flutter speed estimate below 1%, as requested in the assignment.

The structural model was initially solved using a high-fidelity basis ($N_w = 20, N_\theta = 20$) to generate accurate coupled mode shapes. From this full set, a reduced number of coupled modes, denoted as $N_{retained}$, was selected for the aeroelastic stability analysis. The flutter speed was computed repeatedly by increasing $N_{retained}$ (e.g., $N_{retained} = 2$ implies retaining the first two fundamental coupled modes). Figure 2 illustrates the convergence behavior. The analytical reference value for the Goland wing is approximately 137 m/s. The analysis reveals that the flutter mechanism is dominated by the coalescence of the fundamental bending and torsion modes.

Conclusion on Basis Size: Using just the first 2 coupled modes yields a flutter speed of 137.32 m/s. Comparing this to the converged solution (136.99 m/s), the error is approximately 0.24%. Therefore, the minimal set of modes required to achieve an error below 1% is **2 modes**.

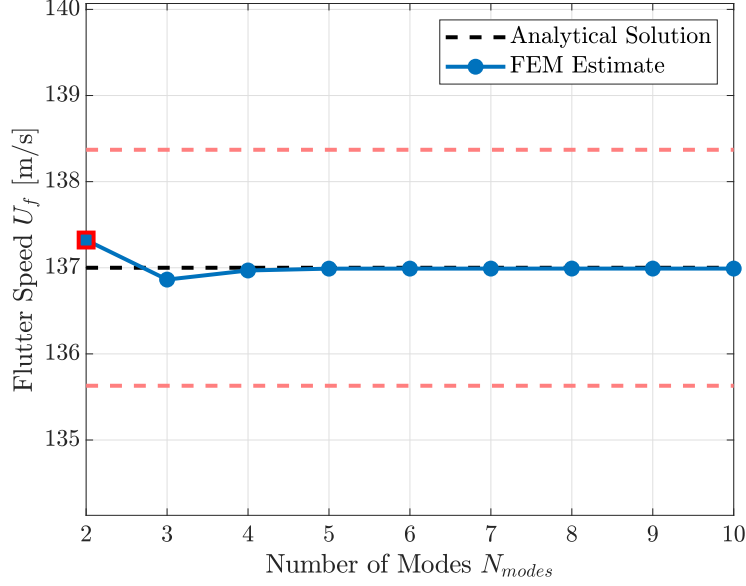


Figure 2: Convergence of the flutter speed estimate as a function of the number of retained coupled modes.

4.3. Stability Results

Using a conservative basis of 10 coupled modes ($N_{retained} = 10$) to ensure robustness for higher-order effects, the final V-g diagram was computed sweeping the velocity from 0 to 200 m/s.

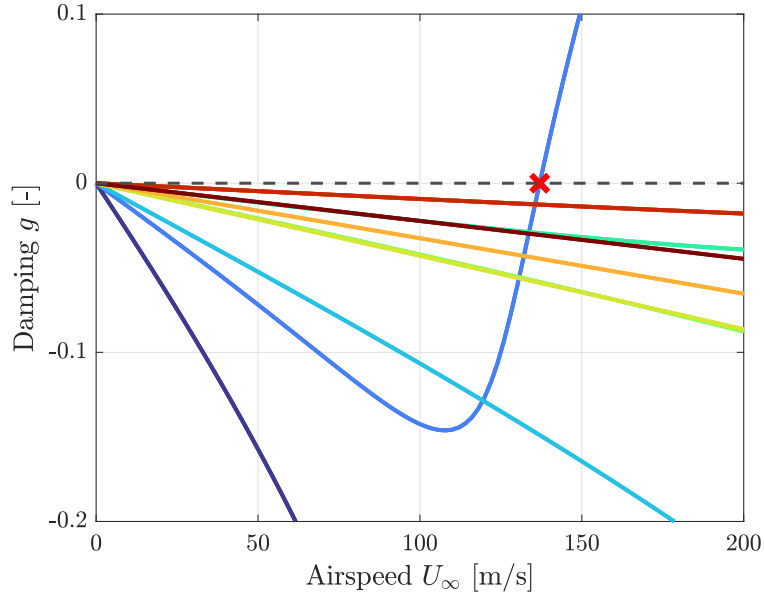


Figure 3: V-g diagram for the Goland Wing showing the damping evolution.

The flutter condition is defined as the speed at which the damping parameter g crosses zero into the positive (unstable) region. The computed critical speed is:

$$U_{flutter} = 136.99 \text{ m/s} \quad (10)$$

This instability corresponds to a classical bending-torsion binary flutter.

5. Task 4: Sensitivity Analysis to Physical Parameters

5.1. Methodology

To assess the robustness of the flutter margins, a sensitivity analysis was performed. The flutter speed U_f was computed while perturbing four key physical parameters: Bending Stiffness (EI), Torsional Stiffness (GJ), Static Unbalance (x_{cg}), and Lift Curve Slope ($C_{L\alpha}$). The variation range was defined from 98% to 102% of the nominal values.

5.2. Results

The computed flutter speeds for the extreme variations ($\pm 2\%$) and the nominal case are summarized in Table 1.

Table 1: Effect of Parameter Variation on Flutter Speed (U_f [m/s])

Condition	EI (Bending)	GJ (Torsion)	x_{cg} (Unbalance)	$C_{L\alpha}$ (Lift Slope)
-2% ($0.98 \times$ Nom)	137.64	134.96	137.37	138.67
Nominal ($1.0 \times$)	136.99	136.99	136.99	136.99
$+2\%$ ($1.02 \times$ Nom)	136.34	139.00	136.64	135.34
Net Change	-1.30 m/s	$+4.04$ m/s	-0.73 m/s	-3.33 m/s

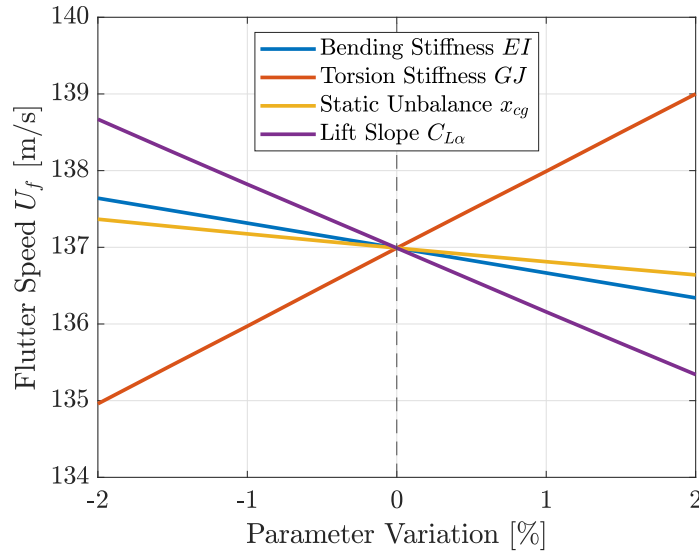


Figure 4: Sensitivity Analysis: Flutter Speed.

5.3. Discussion of Physical Mechanisms

The trends observed in the numerical results (Table 1 and Figure 4) confirm the theoretical expectations for a bending-torsion flutter mechanism:

1. **Bending Stiffness (EI):** Displays a counter-intuitive but physically consistent negative trend. Increasing EI raises ω_h , bringing it closer to ω_α . This slight reduction in frequency separation marginally facilitates the instability, lowering the flutter speed by 1.30 m/s.
2. **Torsional Stiffness (GJ):** This is the dominant stabilizing parameter. Increasing GJ by 2% yields the largest increase in flutter speed (+4.04 m/s). Stiffening the wing in torsion raises the torsional natural frequency ω_α , widening the separation from the bending frequency ω_h and delaying the coalescence.

3. **Static Unbalance (x_{cg}):** Increasing the distance between the Elastic Axis and the Center of Gravity (moving the mass aft) is destabilizing, reducing the flutter speed.
4. **Lift Curve Slope ($C_{L\alpha}$):** Shows a significant negative sensitivity. Increasing the aerodynamic gain by 2% reduces the flutter speed by 3.33 m/s. Higher aerodynamic forces accelerate the energy transfer into the structure.

In conclusion, to improve the flutter margin of the Goland wing, the most effective structural modification is to **increase the torsional stiffness**.

6. Task 5: Effect of Tip Weight and Mass Balancing

6.1. Formulation: Discrete Mass Augmentation

Since the addition of a substantial tip mass ($m_{tip} = 80$ kg, $J_{cg} = 15$ kg m²) significantly alters the dynamic characteristics of the wing, the natural modes of the clean wing are no longer exact solutions. To compute the new coupled modes without re-meshing, a **Modal Synthesis (Rayleigh-Ritz) approach** was employed. The deformation of the loaded wing is approximated as a linear combination of the clean wing's Finite Element mode shapes (Φ). The generalized mass matrix is augmented with the discrete inertia terms projected onto these mode shapes at the wing tip ($x = L$):

$$\mathbf{M}_{total} = \mathbf{M}_{wing} + \Phi(L)^T \mathbf{M}_{tip} \Phi(L) \quad (11)$$

The inertia matrix of the tip weight $\mathbf{M}_{tip}$, expressed in physical coordinates (h, α) at the Elastic Axis, is derived using the parallel axis theorem (Steiner's Theorem):

$$\mathbf{M}_{tip} = \begin{bmatrix} m_{tip} & m_{tip}d_{cg} \\ m_{tip}d_{cg} & J_{cg} + m_{tip}d_{cg}^2 \end{bmatrix} \quad (12)$$

where d_{cg} is the distance from the Elastic Axis to the Center of Gravity of the tip mass (positive aft). The off-diagonal term $S_{tip} = m_{tip}d_{cg}$ represents the static unbalance introduced by the weight, which is the key parameter for mass balancing.

6.2. Results: Sensitivity to CG Position

The flutter speed was computed for a sweep of tip weight positions ranging from the Leading Edge to the Trailing Edge. The specific cases requested in the assignment yielded the results summarized in Table 2.

Table 2: Flutter Speed Results for Tip Weight Configurations

Configuration	CG Position (% Chord)	$U_{flutter}$ (m/s)	Trend
Case A (Elastic Axis)	33%	173.38	Reference
Case B (Forward)	5%	616.23	Highly Stable
Case C (Aft)	50%	137.84	Destabilizing

The results in Figure 5 illustrate the power of mass balancing. Placing the heavy tip mass near the leading edge (Case B) virtually eliminates the flutter phenomenon within the realistic flight envelope ($U_f > 600$ m/s). Consequently, the critical failure mode shifts from Flutter to static divergence. Conversely, moving the mass aft (Case C) negates the inertial stabilization, returning the flutter speed to near clean-wing values.

The target speed is 199.4 m/s, and the sweep analysis confirms that this value is easily achievable. Positioning the tip weight between 22% and 28% of the chord, or from 13% onwards, is shown to be a highly effective strategy.

The discontinuity in the curve represents a transition in the critical instability mechanism. As the mass moves aft, the stabilization diminishes, and the critical mode switches.

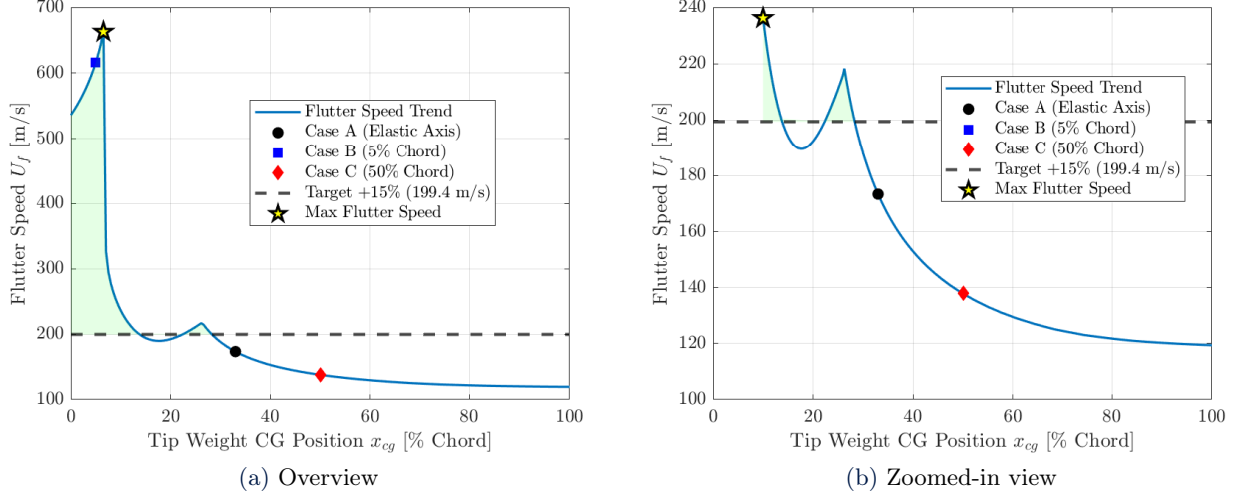


Figure 5: Flutter speed as a function of the Tip Weight CG position. The green region indicates the design space satisfying the +15% requirement.

7. Optional Task 1: Mach Number Effects

7.1. Methodology: Prandtl-Glauert Correction

The baseline flutter analysis assumed incompressible flow ($M \approx 0$). To account for compressibility effects in the subsonic regime, the aerodynamic model was modified using the Prandtl-Glauert correction factor $\beta = \sqrt{1 - M^2}$. The lift curve slope $C_{L\alpha}$, which linearly scales the magnitude of the unsteady aerodynamic forces, is corrected as follows:

$$C_{L\alpha}(M) = \frac{C_{L\alpha, M=0}}{\sqrt{1 - M^2}} = \frac{2\pi}{\sqrt{1 - M^2}} \quad (13)$$

The flutter speed was computed for a range of Mach numbers from $M = 0$ to $M = 0.7$, updating the aerodynamic matrices at each step.

7.2. Results

The variation of the flutter speed with Mach number is presented in Figure 6.

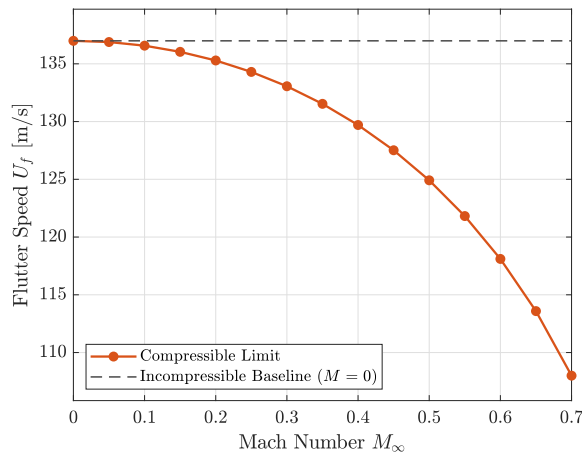


Figure 6: Decrease in flutter speed due to compressibility effects (Prandtl-Glauert correction).

As expected from the sensitivity analysis (Task 4), the increase in the lift slope due to compressibility acts as a destabilizing factor. The flutter speed decreases monotonically as the Mach number increases. For instance, at $M = 0.7$, the effective lift slope increases by a factor of $1/\sqrt{1-0.7^2} \approx 1.4$, causing a significant reduction in the stability margin compared to the incompressible baseline.

8. Optional Task 2: Aileron Effects (Flutter and Reversal)

8.1. Methodology: Analytical Rayleigh-Ritz Formulation

To rigorously analyze the aeroelastic stability of the wing-aileron system, a dedicated Rayleigh-Ritz model was implemented from first principles. Instead of relying on a finite element discretization, analytical shape functions satisfying the geometric boundary conditions (cantilever) were employed for the entire span L :

$$\psi_n(y) = 1 - \cos\left(\frac{n\pi y}{2L}\right) \quad (\text{Bending}), \quad \phi_n(y) = \sin\left(\frac{n\pi y}{2L}\right) \quad (\text{Torsion}) \quad (14)$$

The aileron deflection β was modeled as a discrete degree of freedom, kinematically defined as a unit step function active only over the control surface span $\mathcal{D}_{ail} = [3.35, 5.15]$ m.

8.1.1 Structural Matrices Assembly

The generalized mass ($\mathbf{M}$) and stiffness ($\mathbf{K}$) matrices were assembled by integrating the kinetic and potential energy contributions. A key feature of this implementation is the **zonal integration** of the inertial properties. The mass matrix components are computed as:

$$M_{ij} = \int_0^L (\dots)_{wing} dy + \int_{\mathcal{D}_{ail}} (\dots)_{aileron} dy \quad (15)$$

Specifically, the coupling terms between the wing modes and the aileron degree of freedom are derived by integrating the distributed static unbalance (S_β) and coupled inertia over the aileron span:

$$M_{w\beta}^{(i)} = - \int_{\mathcal{D}_{ail}} \psi_i(y) \cdot S_\beta dy, \quad M_{\theta\beta}^{(j)} = \int_{\mathcal{D}_{ail}} \phi_j(y) \cdot I_{\alpha\beta} dy \quad (16)$$

Here, S_β represents the static unbalance of the aileron about its hinge line. The problem statement specifies that the flap CG is at 50% of the flap chord, placing it significantly aft of the hinge, which results in a large destabilizing unbalance.

8.1.2 Aerodynamic Model (3-DOF Galerkin)

The unsteady aerodynamic forces were modeled using the full 3-degree-of-freedom Theodorsen theory. The local aerodynamic matrices ($\mathbf{K}_{aero}$, $\mathbf{C}_{aero}$, $\mathbf{M}_{aero}$) were constructed using the geometric coefficients $T_1 \dots T_{14}$ derived from the flap geometry ($c = 0.508$, $e = -0.34$). These local matrices were projected onto the global generalized coordinates using Galerkin integrals weighted by the shape functions.

8.2. Solution Methods

8.2.1 Flutter Analysis: p-k Method

The complex eigenvalue problem was solved iteratively:

$$\left[\mathbf{M}p^2 + \mathbf{K} - \frac{1}{2}\rho U^2 \mathbf{Q}(k) \right] \mathbf{q} = 0, \quad p = \gamma + i\omega, \quad k = \frac{\omega b}{U} \quad (17)$$

Eigenvalue tracking across velocities identified the flutter point where $\gamma > 0$.

8.2.2 Control Reversal: Static Aeroelasticity

The reversal speed was computed from static equilibrium ($\omega = 0$, $C(k) = 1$):

$$\left[\mathbf{K} - \frac{1}{2} \rho U^2 \mathbf{Q}(0) \right] \mathbf{q} = \mathbf{F}_\beta \delta \quad (18)$$

The net rolling moment $M_{roll}(U) = \int_0^L L(x) \cdot x dx$ was evaluated, with reversal occurring when $M_{roll}(U_{rev}) = 0$.

8.2.3 Divergence: Static Aeroelasticity

The condition for divergence is expressed mathematically as:

$$\det [\mathbf{K} - q_\infty \mathbf{Q}(0)] = 0 \quad (19)$$

8.3. Results

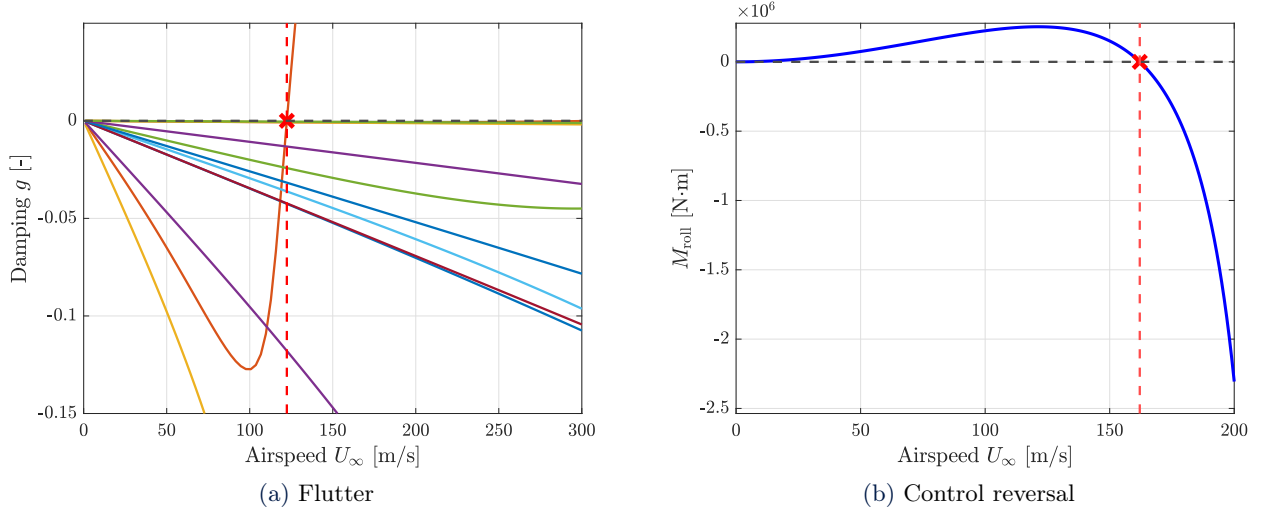


Figure 7: Flutter and Reversal

Phenomenon	Speed [m/s]	Limiting
Clean Wing Flutter	137	—
Wing-Aileron Flutter	122	Yes
Control Reversal	162	No
Divergence	223	No

Table 3: Critical speeds comparison

8.3.1 Physical Interpretation

- **Flutter Reduction:** The aileron's mass and aft CG position (50% chord) create strong inertial coupling, reducing the flutter speed by 11% compared to the clean wing.
- **Control Reversal:** Occurs at 162.0 m/s, well above the flutter speed. The hinge stiffness K_β provides sufficient resistance to maintain control effectiveness throughout the flight envelope. Occurs above flutter speed, thus not limiting.
- **Divergence :** Static torsional instability. The high value indicates substantial torsional stiffness, typical for cantilever wings with moderate aspect ratios.

- **Limiting Phenomenon:** Flutter at 122 m/s remains the critical constraint; control reversal and divergence do not impose additional limitations.

8.4. Conclusion

The aileron-equipped Goland wing exhibits three distinct aeroelastic instabilities, with flutter being the critical limitation at 122 m/s. The results demonstrate that:

- Control surfaces significantly affect dynamic stability but have less impact on static instabilities.
- Proper mass balancing and stiffness distribution can potentially improve the flutter boundary.
- The current configuration provides adequate margins for control reversal and divergence, allowing safe operation up to the flutter speed.